MetaFrontierLab Benchmark Report

Generated from: results/benchmarks_math_upgrade_smoke

Replications per scenario: 1

Methods benchmarked: tbema, exact_baseline, dersimonian_laird, henmi_copas, metafor_trimfill_external, metafor_selmodel_external, copas_selection_external, robma_bibma_external

Executive Summary

- Best overall RMSE in this run: **tbema** at 0.420.
- Highest observed coverage: **tbema** at 0.750.
- Fastest method: **dersimonian_laird** at 0.000 seconds per fit.
- **mean_elapsed_sec** summarizes valid completed fits only; **mean_elapsed_sec_attempted** includes failures and adapter startup overhead.

Overall Method Ranking

ccessful_runs	invalid_ok_runs	skipped_runs	error_runs	success_rate	bias	mean_absolute_error	rmse	coverage_95	mean_ci_width	me
4	0	0	0	1.000	-0.231	0.342	0.420	0.750	1.421	
4	0	0	0	1.000	0.256	0.490	0.587	0.500	0.830	
4	0	0	0	1.000	0.295	0.495	0.599	0.750	0.886	
4	0	0	0	1.000	0.373	0.513	0.608	0.500	0.900	
4	0	0	0	1.000	0.358	0.534	0.636	0.500	0.826	
4	0	0	0	1.000	0.363	0.562	0.652	0.250	1.008	
4	0	0	0	1.000	0.143	0.542	0.657	0.500	1.083	
4	0	0	0	1.000	0.400	0.597	0.668	0.500	1.146	

Scenario Highlights

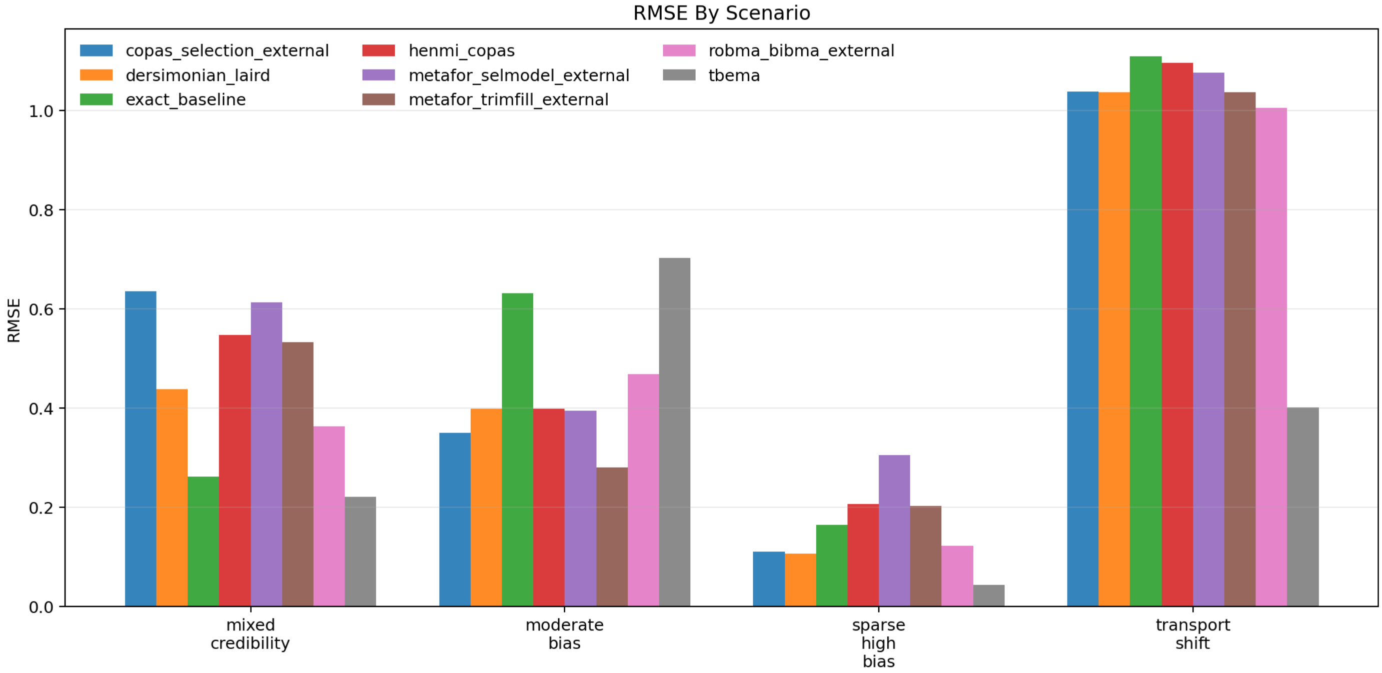
- **mixed_credibility**: best RMSE was tbema (0.221); highest coverage was tbema (1.000).
- **moderate_bias**: best RMSE was metafor_trimfill_external (0.281); highest coverage was metafor_trimfill_external (1.000).
- **sparse_high_bias**: best RMSE was tbema (0.043); highest coverage was tbema (1.000).
- **transport_shift**: best RMSE was tbema (0.401); highest coverage was tbema (1.000).

Scenario Table

eplications_total	successful_runs	invalid_ok_runs	skipped_runs	error_runs	success_rate	bias	rmse	coverage_95	mean_ci_width	mea
1	1	0	0	0	1.000	0.636	0.636	0.000	0.702	
1	1	0	0	0	1.000	0.438	0.438	1.000	0.924	
1	1	0	0	0	1.000	0.262	0.262	1.000	1.243	
1	1	0	0	0	1.000	0.548	0.548	0.000	1.049	
1	1	0	0	0	1.000	0.613	0.613	0.000	1.068	
1	1	0	0	0	1.000	0.533	0.533	0.000	0.980	
1	1	0	0	0	1.000	0.364	0.364	1.000	0.817	
1	1	0	0	0	1.000	0.221	0.221	1.000	0.949	
1	1	0	0	0	1.000	-0.350	0.350	1.000	0.845	
1	1	0	0	0	1.000	-0.399	0.399	1.000	0.824	
1	1	0	0	0	1.000	-0.632	0.632	0.000	1.115	
1	1	0	0	0	1.000	-0.399	0.399	0.000	0.765	
1	1	0	0	0	1.000	-0.394	0.394	1.000	0.829	
1	1	0	0	0	1.000	-0.281	0.281	1.000	0.839	
1	1	0	0	0	1.000	-0.468	0.468	0.000	0.837	
1	1	0	0	0	1.000	-0.702	0.702	0.000	1.254	
1	1	0	0	0	1.000	0.110	0.110	1.000	1.131	
1	1	0	0	0	1.000	0.106	0.106	1.000	1.161	
1	1	0	0	0	1.000	-0.165	0.165	1.000	1.461	
1	1	0	0	0	1.000	0.206	0.206	1.000	1.450	
1	1	0	0	0	1.000	0.305	0.305	1.000	1.692	
1	1	0	0	0	1.000	0.203	0.203	1.000	1.144	
1	1	0	0	0	1.000	0.123	0.123	1.000	0.990	
1	1	0	0	0	1.000	-0.043	0.043	1.000	1.543	
1	1	0	0	0	1.000	1.038	1.038	0.000	0.628	
1	1	0	0	0	1.000	1.036	1.036	0.000	0.636	
1	1	0	0	0	1.000	1.109	1.109	0.000	0.515	
1	1	0	0	0	1.000	1.096	1.096	0.000	0.768	
1	1	0	0	0	1.000	1.076	1.076	0.000	0.997	
1	1	0	0	0	1.000	1.036	1.036	0.000	0.636	
1	1	0	0	0	1.000	1.005	1.005	0.000	0.678	
1	1	0	0	0	1.000	-0.401	0.401	1.000	1.940	

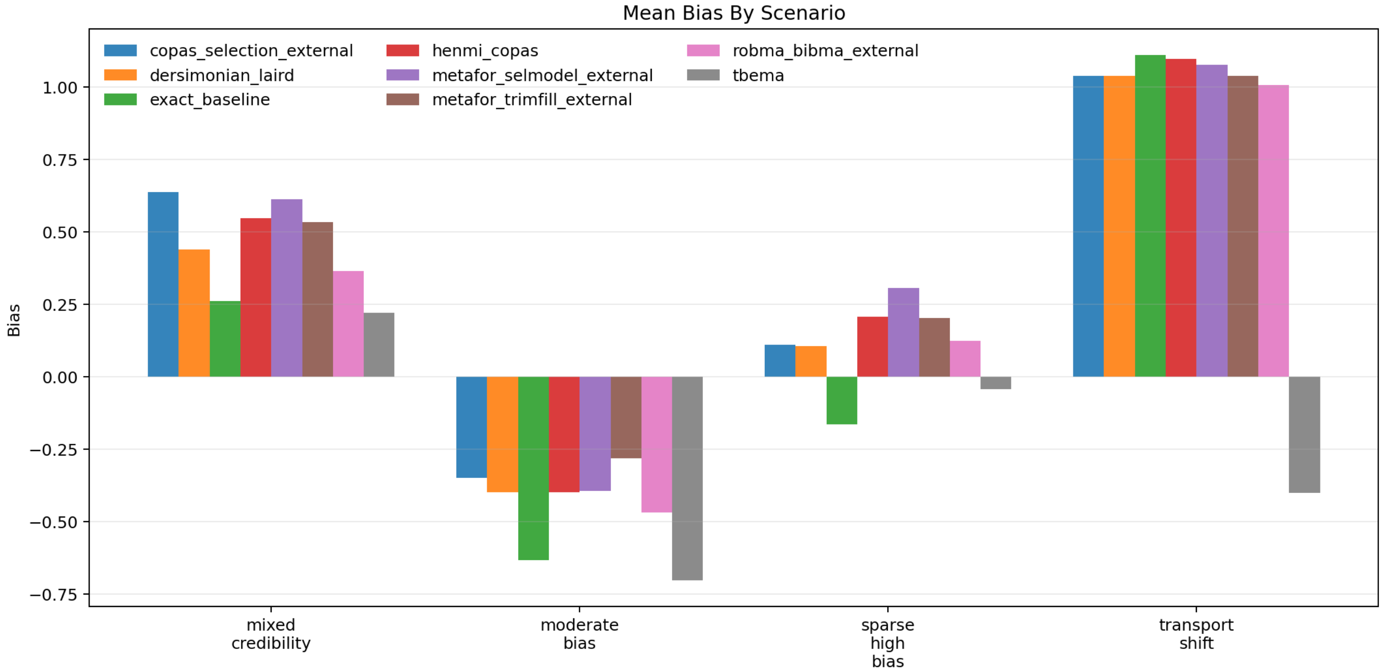
RMSE by Scenario

Figure generated from the benchmark summary.



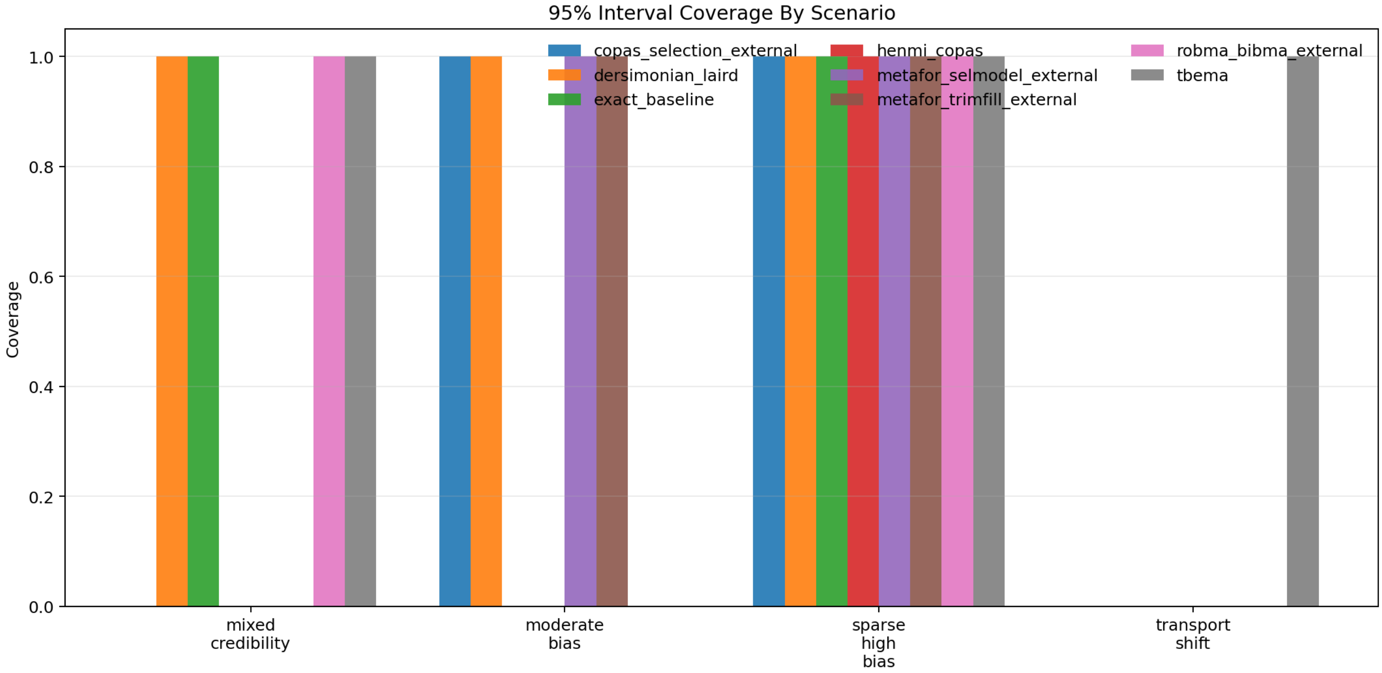
Bias by Scenario

Figure generated from the benchmark summary.



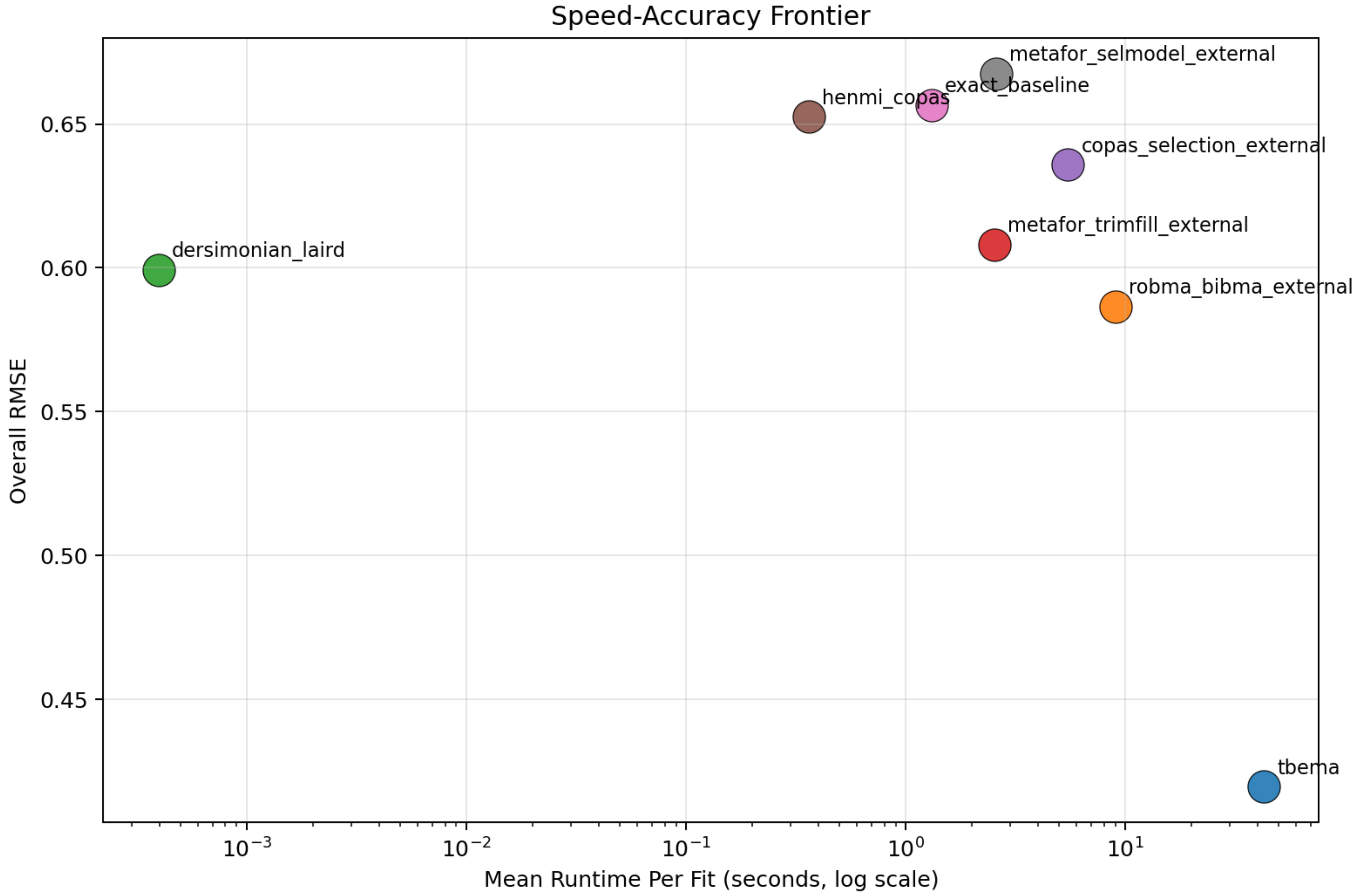
Coverage by Scenario

Figure generated from the benchmark summary.



Speed-Accuracy Frontier

Figure generated from the benchmark summary.



Methods Appendix

- TBEMA: exact sparse-data meta-analysis with stacked bias tempering, transport weighting, and ridge-regularized target-aware meta-regression.
- DerSimonian-Laird: standard normal-normal random-effects benchmark.
- Henmi-Copas: publication-bias-robust interval method.
- metafor trimfill: trim-and-fill adjustment from metafor.
- metafor selmodel: step-function selection model from metafor.
- Copas selection: Copas model from metasens.
- RoBMA BiBMA: Bayesian model averaging for binary outcomes, using the fast spike-and-slab algorithm.